

Mend-Amar Badral

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I'm a diligent and hard-working student with a passion for facing new and exciting challenges. I believe in the power of regular habit and teamwork.

EDUCATION

Budapest University of Technology and Economics (BME) M.S. in Computer Science Engineering <i>Focus: Data Science & Artificial Intelligence</i>	Sep, 2025 – June, 2027 Budapest, Hungary
National University of Mongolia (NUM) B.S. in Applied Mathematics Thesis: Evaluating land degradation using image processing methods	Sep, 2021 – June, 2025 Ulaanbaatar, Mongolia

EXPERIENCE

Center of Mathematics Application Research Laboratory <i>Research Undergraduate Intern</i>	Oct, 2023 – June, 2025
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- **Lab Leadership:** Selected as Undergraduate Lab Lead in final year; coordinated weekly research progress meetings and facilitated communication between students and faculty.
- **Infrastructure Management:** Managed and maintained the lab computing resources in the Linux environment.
- **Research:** Conducted research on image processing and machine learning techniques to quantify plant types from drone imagery using segmentation methods (thresholding, K-means, Moran's I autocorrelation).
- **Documentation:** Authored a comprehensive laboratory handbook and technical documentation to standardize onboarding for future researchers.

PROJECTS

[Deep Learning Foundations and Concepts Book Figures Reproduction](#)

- Self-initiated project to reproduce key examples from C.Bishop and H.Bishop's *Deep Learning Foundations and Concepts* to gain theoretical and fundamental understanding of polynomial regression and probability theory based on **Monte Carlo Simulation**.
- Published reproducible Jupyter and Pluto.jl notebooks on GitHub, with emphasis on clear explanation, interactivity of plots, figures, and visualization.

SKILLS

Programming Languages:	Python, JavaScript, Julia, R, C, C++
ML Frameworks:	PyTorch, JAX
Markup & Docs:	L ^A T _E X
Databases & Tools:	Git, SQLite, MySQL
Theoretical Background:	Applied Mathematics, Statistics, Machine Learning